

llmXive Automated Review of BlockPilot: Instance-Adaptive Policy Learning for Diffusion-based Speculative Decoding

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer’s exact instructions are public in the [llmXive agent-prompt library](#); each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer’s section also names the underlying model that produced it.

This paper’s panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_claim_accuracy.md

The paper’s central claims regarding the variability of optimal block size, the locality of this distribution, and the performance gains of BlockPilot are well-supported by the internal evidence presented in the tables. The numerical claims in the abstract ($4.20\times$ speedup, 5.92 acceptance length) match the corresponding entries in Table 1 for Qwen3-4B at T=1 exactly. The comparison with baselines (DFlash, EAGLE-3) is consistent with the data in Table 1 and Table 2.

However, there is a missing citation entry in the bibliography. Section 4.1 and Appendix Table 2 reference the model “Qwen3-Coder-30B-A3B” with the citation key `cao2026qwen3`. The provided bibliography file (`neurips_2025.bib`) does not contain an entry for this key. While the model itself may exist in the authors’ private repository or a future release (consistent with the 2026 date), the reference is technically incomplete in the provided source, preventing a reader from verifying the source of this model or its specifications. This is a minor issue fixable by adding the missing bib entry.

Additionally, the paper cites `chen2026dflash` and `cao2026qwen3` with 2026 dates. Given the paper’s own arXiv ID (2606.31315) and the context of a 2026 preprint, these dates are internally consistent, though they represent future-dated works relative to the current real-world date. This is not an accuracy error within the context of the paper’s timeline.

No other claim-accuracy mismatches, overstatements, or unsupported assertions were found. The logic connecting the “locality” finding to the “classification” formulation is sound and supported by the ablation studies.

Action items.

- **[writing]** Section 4.1 cites ‘cao2026qwen3’ for Qwen3-Coder-30B-A3B, but this key is missing from `neurips_2025.bib`. Add the entry or remove the citation to ensure reproducibility. (writing)

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_figure_critic.md

Figure 1

Figure 1 presents a clear bar chart comparing speedups across models, but the legend entry ‘Ours’ is undefined in the caption, the y-axis lacks baseline context, and no uncertainty measures are provided.

Figure 2

Figure 2 presents a clear bar chart comparing proportions, but the legend’s connection to the bar colors is implicit rather than explicit, and the caption could be more precise in defining what each color represents to avoid ambiguity.

Figure 3

Figure 3 provides a clear and well-structured schematic of the BlockPilot inference pipeline. The visual flow from input sequences through the LLM Target Model, Block Predictor, and dLLM Draft Model is logical and easy to follow. The caption accurately describes the process shown, and all components are clearly labeled.

Action items.

- **[writing]** Figure 1: The legend entry ‘Ours’ is not defined in the caption; the caption only defines ‘DFlash(n)’ but does not explicitly name the proposed method.
- **[writing]** Figure 1: The y-axis label ‘SpeedUp ($\times$)’ is ambiguous; it should clarify whether this is relative to a baseline (e.g., ‘relative to EAGLE-3’ or ‘vs. baseline’).
- **[science]** Figure 1: No error bars or confidence intervals are shown despite multiple models and settings; this limits assessment of statistical significance of the reported speedups.
- **[science]** Figure 2: The y-axis is labeled ‘Proportion’ (0 to 1), but the bars for each group (e.g., GSM8K: $0.34 + 0.66 = 1.0$) sum to 1.0, implying they represent the full distribution of a binary outcome. However, the caption states ‘Proportion of samples with B^* matching or mismatching B ’, which is ambiguous. It is unclear if the blue bars represent ‘matching’ and orange ‘mismatching’, or vice versa. The legend at the top defines $B^* = B$ (blue) and $B^* \neq B$ (orange), but this legend is not clearly linked to
- **[writing]** Figure 2: The legend at the top uses colored squares to denote $B^* = B$ (blue) and $B^* \neq B$ (orange), but these colors are not explicitly stated in the caption. While the colors match the bars, the caption should clarify that blue represents ‘ $B^* = B$ ’ and orange represents ‘ $B^* \neq B$ ’ for readers who may not immediately associate the legend with the bars.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: qwen.qwen3.5-122b

Prompt: `agents/prompts/paper_reviewer_jargon_police.md`

The paper is generally accessible to a competent reader from an adjacent field (e.g., a researcher in NLP or systems familiar with LLMs but not specifically with diffusion-based speculative decoding). Most core concepts like “speculative decoding,” “block size,” and “prefilling” are standard or well-explained in context. However, there are a few instances of undefined acronyms and overloaded symbols that create minor friction.

Specifically, the acronym “dLLM” is used in the Introduction and Figure captions before being explicitly expanded in the text. While the full phrase “Diffusion-based language models” appears nearby, the acronym itself should be defined at its first occurrence to prevent a reader from having to guess or search for the expansion. Additionally, the symbol $\mathcal{B}$ is used in the problem formulation equations before its definition as the candidate set is provided in the subsequent subsection. While the context makes the meaning clear, strict adherence to self-containment requires defining the set before using it in equations.

There is also a potential ambiguity with the symbol k . It is used in Section 2.2 to denote the “Top-k probabilities” (a ranking concept) and later as the “candidate interval radius” (a range parameter). While the context distinguishes them, an adjacent-field reader might momentarily confuse the two usages. A brief clarification distinguishing the “Top-k” selection from the interval radius k would eliminate this potential confusion. Finally, the model name “Qwen3-Coder-30B-A3B” contains a suffix “A3B” that is not standard nomenclature; a brief parenthetical explanation of what this suffix denotes (e.g., a specific architecture variant) would aid clarity.

These are minor issues that can be resolved with simple text edits (adding parenthetical expansions or clarifying clauses) and do not require re-running experiments or changing the scientific content.

Action items.

- **[writing]** Section 1 (Introduction) and Abstract use ‘dLLM’ and ‘dLLMs’ without defining the acronym. While ‘Diffusion-based language models’ is spelled out in the text, the acronym ‘dLLM’ appears in Figure 1 caption and Section 1 before being explicitly defined as ‘Diffusion-based Language Models (dLLMs)’. Add the expansion at first use.
- **[writing]** Section 2.1 (Problem Formulation) introduces the symbol $au(\mathcal{B})$ as ‘expected number of accepted tokens’ but does not define the set $\mathcal{B}$ until Section 2.2. While $\mathcal{B}$ is defined shortly after, the symbol appears in Eq 1 and Eq 3 before its definition. Define $\mathcal{B}$ immediately before or within the first equation where it is used.
- **[writing]** Section 2.2 (Key Findings) uses the term ‘Top-k probabilities’ in the context of overfitting. While ‘k’ is later defined as the interval radius, the ‘Top-k’ here refers to a different concept (ranking probabilities). This creates ambiguity for the symbol ‘k’. Clarify that ‘Top-k’ refers to the k highest probability tokens, distinct from the interval radius parameter k .
- **[writing]** Section 3.1 (Experimental Setup) lists ‘Qwen3-Coder-30B-A3B’ and ‘Qwen3-4B’ without defining the ‘A3B’ suffix or the specific architecture variant. For an adjacent-field

reader, ‘A3B’ is opaque. Add a brief gloss (e.g., ‘...Qwen3-Coder-30B-A3B (a specific MoE variant)’)) or ensure the citation [cao2026qwen3] is sufficient context, but a one-sentence gloss is safer.

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_logical_consistency.md

The paper presents a coherent argument for adaptive block size selection, but contains a logical inconsistency between the theoretical justification and the experimental configuration regarding the candidate interval radius k .

In Section 2.2 (“Key Findings”, Finding II), the authors state that the range $[B - 3, B + 3]$ (i.e., $k = 3$) “covers the optimal size for nearly all samples.” This premise suggests that a search space of $k = 3$ is necessary to capture the optimal block size. However, the methodology in Section 2.3 and the implementation details in Section 4.1 explicitly state that the predictor is trained and evaluated with $k = 2$. Furthermore, the ablation study in Table 4 demonstrates that $k = 2$ yields the highest speedup (4.76x), outperforming $k = 3$ (4.64x).

The argument is broken because the premise (that $k = 3$ is required for coverage) does not logically support the conclusion (that $k = 2$ is the optimal setting). If $k = 3$ is indeed required to cover nearly all optimal cases, restricting the search to $k = 2$ should theoretically incur a performance penalty due to missed optima, yet the results show the opposite. The text in Section 2.2 must be revised to align with the empirical evidence, either by clarifying that $k = 2$ is sufficient for the specific datasets or by explaining why the theoretical bound of $k = 3$ does not translate to better performance (e.g., noise in the tail of the distribution). As written, the paper asserts a necessity for $k = 3$ while empirically demonstrating the superiority of $k = 2$ without resolving this contradiction.

Action items.

- **[writing]** Section 2.2 claims $k=3$ covers ‘nearly all cases’ for optimal block size, yet Section 2.3 and Table 4 show $k=2$ is used and performs better. Reconcile the theoretical claim with the empirical choice of $k=2$.

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_overreach.md

The paper generally aligns its core methodological claims with the provided evidence. However,

three rhetorical overreaches require correction:

1. **SOTA and “All Settings” Claims:** The abstract and Figure 1 caption assert “state-of-the-art” performance and “highest acceleration across all settings.” The experiments (Table 1) only compare BlockPilot against EAGLE-3 and DFlash on Qwen3-4B/8B and Llama-3.1-8B. The claim of SOTA status is not licensed without comparisons to other diffusion-based speculative decoding baselines (e.g., DiffuSpec, SpecDiff-2) or a broader range of model architectures. The phrase “all settings” is particularly strong given the limited model and temperature range tested.
2. **“Plug-and-Play” Framing:** The abstract describes the method as “plug-and-play,” and the conclusion states it “integrates seamlessly.” While the *inference* overhead is minimal (7.34ms), the *training* phase requires an expensive offline data construction process involving exhaustive sweeps (Appendix Limitations: ~25 seconds per sample for a 32B model). The current rhetoric obscures this significant upfront cost, potentially misleading readers about the ease of adoption. The “plug-and-play” claim should be qualified to apply only to the inference stage.
3. **Universality of Locality:** The paper claims the optimal block size distribution has “strong locality” covering “nearly all samples” within a narrow interval (Section 2.2). This finding is derived from ShareGPT, WSC, and COPA. The language implies this property holds universally for any input, whereas the evidence is limited to these specific datasets. The claim should be scoped to the evaluated domains to avoid overgeneralization.

These issues are primarily rhetorical and can be resolved by tightening the language in the abstract and conclusion to match the specific boundaries of the experimental evidence.

Action items.

- **[writing]** Abstract claims ‘SOTA’ and ‘highest acceleration across all settings’ (Fig 1). Table 1 only compares EAGLE-3/DFlash on Qwen3/Llama. Narrow to ‘outperforms DFlash/EAGLE-3 on tested configs’ or add missing baselines.
- **[writing]** Abstract calls method ‘plug-and-play’ but Appendix Limitations notes ~25s/sample offline data construction cost. Clarify ‘plug-and-play’ applies only to inference, not training, to avoid misleading adoption claims.
- **[writing]** Section 2.2 claims locality covers ‘nearly all samples’ based on ShareGPT/WSC/-COPA. Scope this to ‘observed on evaluated datasets’ as universality across arbitrary domains is untested.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_safety_ethics.md

This work presents a method for adaptive block size selection in diffusion-based speculative decoding, a technique focused on inference efficiency. The methodology involves training a lightweight classifier on public benchmark datasets (ShareGPT, WSC, COPA) and standard model checkpoints (Qwen3, Llama-3.1).

A review of the safety and ethics lens reveals no foreseeable, non-trivial risks of harm that are unaddressed:

1. **Dual-Use Capability:** The method optimizes inference speed (latency reduction) and does not introduce new capabilities for generating harmful content, bypassing safety filters, or conducting cyberattacks. It is a performance optimization layer for existing models.
2. **Data Provenance & Privacy:** The datasets cited (ShareGPT, WSC, COPA) are standard public benchmarks. The paper does not claim to use private, sensitive, or personally identifiable information (PII), nor does it release a new dataset containing raw user data. The “Supervised Data Construction” described is an offline enumeration of block sizes on these public sets, not a collection of new human-subject data.
3. **Human Subjects:** The NeurIPS checklist correctly identifies that no human subjects research was conducted, and no IRB approval is required for the use of these public, pre-existing datasets.
4. **Safeguards:** The “Safeguards” and “Broader Impacts” sections in the checklist appropriately note the absence of high-risk datasets or models. The work does not involve systems designed to deceive, manipulate, or surveil.

The paper is a standard systems/efficiency contribution with no specific, nameable safety gaps requiring mitigation or disclosure beyond what is already present. The verdict is **accept** with no action items.

Scientific Evidence

Weights the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_scientific_evidence.md

The paper presents a compelling hypothesis that optimal block sizes in diffusion-based speculative decoding vary by instance but exhibit locality. However, the experimental design currently lacks the rigor required to support the headline claims of instance-adaptive superiority and stability.

First, the primary results in Table 1 are presented as single-point estimates (e.g., 4.20x speedup) without any measure of variance, standard deviation, or seed count. In speculative decoding, the acceptance length τ is a stochastic variable heavily influenced by the randomness of the draft model and the input distribution. A single run is insufficient to distinguish a genuine methodological improvement from lucky sampling or noise. The authors must report results averaged over multiple random seeds (at least 3-5) with standard deviations or confidence

intervals to demonstrate that the observed gains are stable and reproducible.

Second, the comparison against baselines is potentially confounded. The paper compares BlockPilot against DFlash with fixed block sizes (4, 8, 16, 32). While BlockPilot outperforms these specific fixed settings, the paper does not explicitly compare against the “oracle” fixed block size—the single best n found via the exhaustive sweep described in Section 3.2. If the optimal fixed size (e.g., $n = 16$) is very close to the average performance of BlockPilot, the claim of “instance-adaptive” superiority is weakened. The design needs a direct comparison to the best-performing fixed baseline to isolate the value of the adaptive policy.

Finally, the ablation studies (Section 4.3) vary the predictor’s architecture and search radius k but omit a critical control: a “fixed policy” ablation where the predictor is forced to always output the training block size B . Without this, it is unclear if the predictor is truly learning to adapt to difficult instances or if it is simply learning to output B (the mode of the distribution) and the gains are merely due to the overhead of the predictor being negligible. A control run with a fixed block size equal to the training configuration is necessary to prove the adaptive component is driving the performance.

Action items.

- **[writing]** The paper presents a compelling hypothesis that optimal block sizes in diffusion-based speculative decoding vary by instance but exhibit locality. However, the experimental design currently lacks the rigor required to support the headline claims of instance-adaptive superiority and stability. First, the primary results in Table 1 are presented as single-point estimates (e.g., 4.20x speedup) without any measure of variance, standard deviation, or seed count. In speculative decoding, the acceptance

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_statistical_analysis.md

The statistical treatment of the results in this paper is insufficient to support the quantitative claims made. The primary issue is the complete absence of uncertainty reporting. The authors present specific speedup ratios (e.g., 4.20x) and acceptance lengths (e.g., 5.92) as fixed properties of the method, yet the NeurIPS checklist explicitly states, “We do not report error bars.” In the context of deep learning, where results vary significantly across random seeds, training runs, and hardware conditions, a single point estimate is statistically meaningless. It is impossible for a reader to judge whether the reported improvements over baselines (e.g., 4.17x vs 3.99x) are real effects or simply noise.

Furthermore, the paper uses language implying statistical significance (“consistent speedup gains,” “achieves the best performance”) without providing the necessary inferential machinery (p-values, confidence intervals, or effect sizes with variance). The ablation studies suffer from the same flaw; selecting the optimal hyperparameters based solely on point estimates without

variance data invites overfitting to a specific random seed. While the field often accepts mean-over-seeds without formal p-values, the total absence of any spread metric (SD, SE, or range) is a critical reporting failure. The authors must re-run experiments with multiple seeds to generate variance data or explicitly acknowledge that the reported numbers are from a single run and should not be interpreted as definitive performance metrics.

Action items.

- **[writing]** The statistical treatment of the results in this paper is insufficient to support the quantitative claims made. The primary issue is the complete absence of uncertainty reporting. The authors present specific speedup ratios (e.g., 4.20x) and acceptance lengths (e.g., 5.92) as fixed properties of the method, yet the NeurIPS checklist explicitly states, “We do not report error bars.” In the context of deep learning, where results vary significantly across random seeds, training runs, and hardware

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_writing_quality.md

The paper is generally well-structured, with a clear logical flow from problem formulation to methodology and results. The abstract effectively summarizes the core contribution, and the introduction successfully sets up the motivation for adaptive block sizes. However, there are specific instances where clarity is compromised by ambiguous variable naming, telegraphic phrasing, and missing transitions.

In Section 2.2 (Finding III), the authors discuss using “Top-k probabilities” as an input feature. This creates immediate confusion because the symbol k was previously defined in Finding II as the radius of the candidate interval ($B \pm k$). While a careful reader might infer these are different variables, the reuse of the same symbol for two distinct concepts in close proximity forces the reader to backtrack and re-parse the context. This is a significant friction point that should be resolved by renaming one of the variables (e.g., using m for the Top- m probabilities) or explicitly distinguishing them in the text.

The “Implementation Details” paragraph in Section 3.1 suffers from a lack of grammatical completeness. Phrases like “100 epochs, Adam, $1e^{-5}$ ” read like a checklist rather than a narrative description of the experimental setup. While common in some technical contexts, in a formal paper, these should be woven into complete sentences to maintain a professional tone and ensure the text flows smoothly without requiring the reader to mentally fill in the missing verbs and subjects.

Additionally, the transition between the general claim of “similar gains” and the specific data points in Section 3.2 (Main Results) is abrupt. The text states that gains hold for Qwen3-8B and then immediately lists numbers. A brief bridging sentence explicitly stating that the Qwen3-8B results follow the same trend as the Qwen3-4B results would improve the narrative cohesion.

Finally, the theoretical analysis in the appendix introduces the retention function $r(b; B)$ with a formula but lacks a preceding sentence that intuitively explains what this function represents (i.e., the penalty for deviating from the training block size). Adding this conceptual lead-in would help the reader understand the motivation behind the equation before encountering the mathematical notation.

Addressing these specific points will significantly reduce the cognitive load on the reader and ensure the paper’s high-quality science is delivered with equally high-quality prose.

Action items.

- **[writing]** The paper is generally well-structured, with a clear logical flow from problem formulation to methodology and results. The abstract effectively summarizes the core contribution, and the introduction successfully sets up the motivation for adaptive block sizes. However, there are specific instances where clarity is compromised by ambiguous variable naming, telegraphic phrasing, and missing transitions. In Section 2.2 (Finding III), the authors discuss using “Top-k probabilities” as an input featu